

Chapter 3

Strategies and the Normal Form

The most important concept in game theory is the notion of a strategy. Once we understand strategies, we can describe any game in a second, more compact way: the normal form.

We will revisit the Katzenberg–Eisner game from last week to see how the two forms connect, then meet several classic games that recur throughout the course.

What Is a Strategy?

Definition. A **strategy** is a complete contingent plan for a player in the game.

The word “complete” is what makes this definition powerful, and occasionally confusing. A strategy is not just what a player intends to do; it is a full specification of what she would do at *every* one of her information sets, including ones her own plan might never actually reach.

If a player has three information sets, her strategy names one action at each of the three, even if choosing a particular action at the first would make the other two irrelevant to how the game actually unfolds.

Why “Complete” Matters

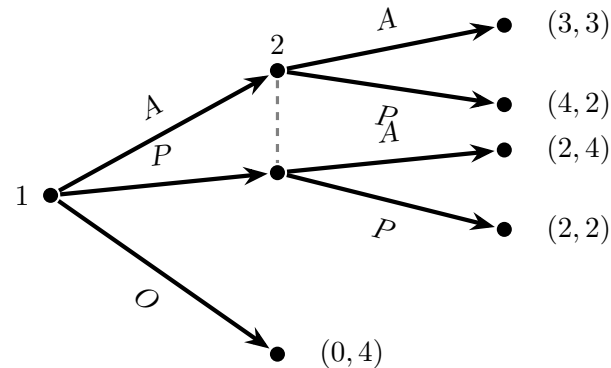
Suppose you are advising a player by mail. You write your instructions on a slip of paper and leave before the game begins, so you cannot react to anything that happens.

Writing “choose Stay” is not enough guidance, even if you are confident that Stay is correct. If the player makes a mistake, or if we simply want to evaluate the plan at every point in the tree, we need instructions for every information set, not just the first one reached.

This is also why we need well-chosen labels: a strategy is written by stringing together the action chosen at each information set, so two different information sets must never share an action label.

Example: An Entry Game

Firm 1 decides whether to compete Aggressively (A), compete Passively (P), or leave the market (Out, O). If firm 1 stays in, firm 2 decides whether to compete Aggressively or Passively, but firm 2 only observes that firm 1 stayed in, not which stance it took.

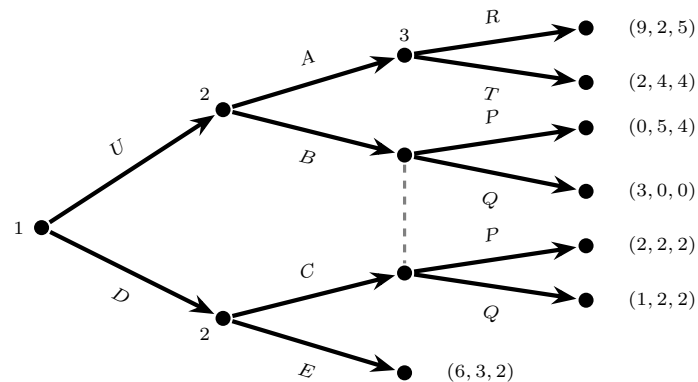


Firm 1 has one information set, so $S_1 = \{A, P, O\}$. Firm 2's two arrival nodes share one information set, so $S_2 = \{A, P\}$ as well, just two strategies despite the game having four possible ways to reach a terminal node.

Discussion: A Three-Player Game

Player 1 chooses U or D , giving player 2 two separate information sets, one per branch. Two of player 3's possible arrival nodes go further and share a single information set.

Before the next slide, work out S_1 , S_2 , and S_3 for yourself.



Answer: A Three-Player Game

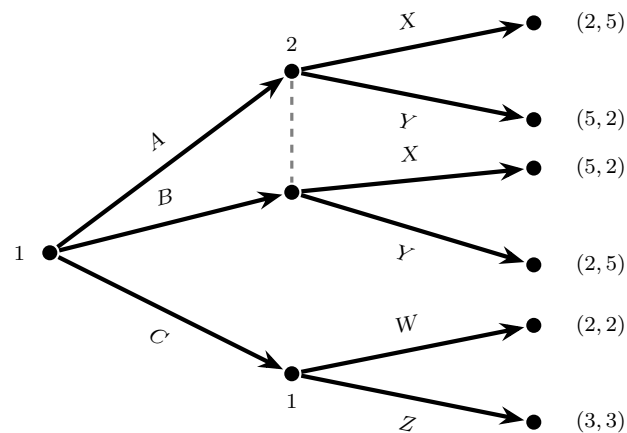
Player 1 has one information set, so $S_1 = \{U, D\}$.

Player 2 has two separate information sets, one reached by U and one by D , so a strategy names an action at each: $S_2 = \{AC, AE, BC, BE\}$.

Player 3 combines one action from his shared $\{B, C\}$ -node with one from his A -node: $S_3 = \{RP, RQ, TP, TQ\}$.

Discussion: Contingent Plans, Again

Player 1 chooses among A , B , and C . Actions A and B both lead to player 2's single information set; action C instead leads player 1 herself back to a second, singleton information set.



Before the next slide, work out S_1 and S_2 for yourself.

Answer: Contingent Plans, Again

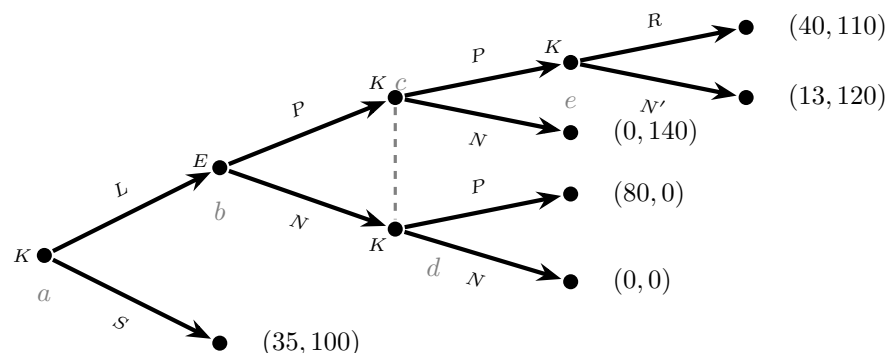
Player 1's plan must specify W or Z even for a strategy that starts with A or B and so never reaches that second information set, six strategies from just three first-move actions:

$$S_1 = \{AW, BW, CW, AZ, BZ, CZ\}.$$

Player 2 has a single information set, so $S_2 = \{X, Y\}$.

Example: Katzenberg's Strategy Space

Recall Katzenberg's three information sets from the Bug's Life game: node a (Leave or Stay), the $\{c, d\}$ information set (Produce *Antz* or Not), and node e (Release early or Not).



A strategy names one action at each information set. For example, LNR means Leave, then Not produce *Antz*, then, if that later information set were somehow reached anyway, Release early.

Katzenberg's Strategies

Note the third information set's actions are labeled R and N' , not R and N : since the second information set already uses N , reusing it here would make a strategy like “ LNN ” ambiguous about which N is which.

Since Katzenberg has two choices at each of three information sets, he has $2 \times 2 \times 2 = 8$ strategies in total:

$$S_K = \{LPR, LPN', LNR, LNN', SPR, SPN', SNR, SNN'\}.$$

Eisner, who has only one information set, has just two: $S_E = \{P, N\}$.

Notation for Strategies

We write S_i for player i 's **strategy space**: the set of all strategies available to her. A generic strategy for player i is $s_i \in S_i$.

Definition. A **strategy profile** is a vector $s = (s_1, s_2, \dots, s_n)$ listing one strategy for every player. The set of all strategy profiles is $S = S_1 \times S_2 \times \dots \times S_n$.

We write $-i$ for “every player except i ,” so s_{-i} lists everyone's strategy except player i 's, and we can split a profile as $s = (s_i, s_{-i})$.

The Normal Form

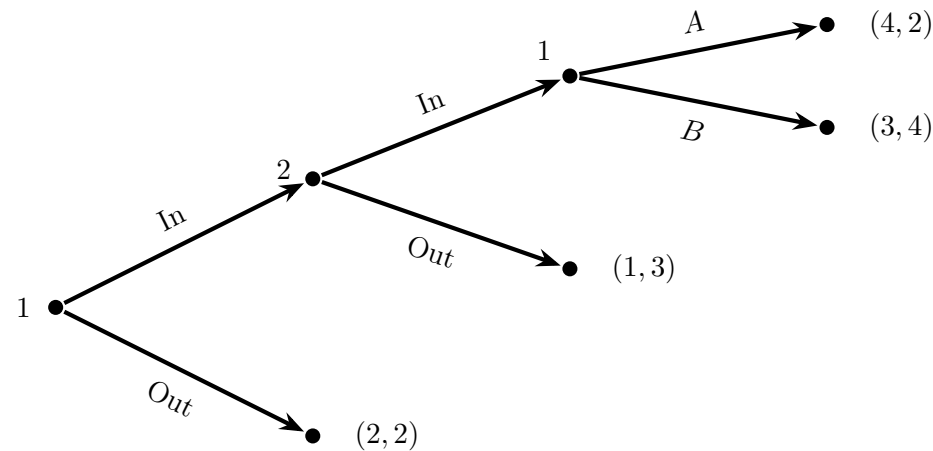
Every strategy profile pins down a path through the tree, and therefore a terminal node and a payoff vector. This lets us define, for each player i , a **payoff function** $u_i : S \rightarrow \mathbb{R}$.

Definition. A game in **normal form** (or **strategic form**) consists of a set of players, strategy spaces $S_1, \dots, S_n$, and payoff functions $u_1, \dots, u_n$.

For two players with finitely many strategies each, we can draw the normal form as a **matrix**: one row per strategy of player 1, one column per strategy of player 2, and player 1's payoff listed first in every cell.

Example: From Tree to Matrix

Consider a simpler game. Player 1 chooses Out or In. If Out, the game ends. If In, player 2 chooses Out or In. If player 2 also chooses In, player 1 makes one more choice, between A and B .



Player 1's second decision only matters along the “In, In” path, but his strategy must still name an action there: $S_1 = \{OA, OB, IA, IB\}$ and $S_2 = \{O, I\}$.

The Resulting Matrix

To fill in each cell, trace the tree using the actions the profile specifies. At (OA, O) , player 1's first move already ends the game at Out, so A is never used and the payoff is $(2, 2)$; likewise for (OB, O) and (OB, I) .

	O	I
OA	2, 2	2, 2
OB	2, 2	2, 2
IA	1, 3	4, 2
IB	1, 3	3, 4

The first two rows coincide: once player 1 plans to choose Out, his unused plan for A versus B cannot affect the outcome.

Guided Exercise: Katzenberg's Matrix

Katzenberg has 8 strategies and Eisner has 2, so the normal form is an 8×2 matrix, filled in by tracing the tree for each profile:

	P	N
LPR	40, 110	80, 0
LPN'	13, 120	80, 0
LNR	0, 140	0, 0
LNN'	0, 140	0, 0
SPR	35, 100	35, 100
SPN'	35, 100	35, 100
SNR	35, 100	35, 100
SNN'	35, 100	35, 100

All four “ S ___” rows coincide, since Stay ends the game early.

Classic Normal-Form Games

These small games recur throughout the course. In each, player 1 picks a row and player 2 picks a column, simultaneously and independently.

	H	T
H	1, -1	-1, 1
T	-1, 1	1, -1

Matching Pennies

	C	D
C	2, 2	0, 3
D	3, 0	1, 1

Prisoners' Dilemma

	Opera	Movie
Opera	2, 1	0, 0
Movie	0, 0	1, 2

Battle of the Sexes

	H	D
H	0, 0	3, 1
D	1, 3	2, 2

Chicken

	A	B
A	1, 1	0, 0
B	0, 0	1, 1

Coordination

	A	B
A	2, 2	0, 0
B	0, 0	1, 1

Pareto Coordination

Example: The Prisoners' Dilemma

Two suspects are held in separate rooms. The police have enough evidence to convict both of a minor charge, but need a confession to convict either of the serious one.

Each may cooperate with his partner by staying silent (C), or defect by confessing (D). If both stay silent, both get a light sentence. If both confess, both get a moderate sentence. If one confesses and the other stays silent, the confessor goes free and the silent partner is punished severely.

Each prisoner ranks the outcomes: best is confessing while the other stays silent (payoff 3), then mutual silence (2), then mutual confession (1), and worst is staying silent while the other confesses (0).

There is a tension between what is best for each player and what is best for the pair, a recurrent theme in game theory.

Example: The Stag Hunt

Two hunters can each pursue a stag or a hare. Catching the stag requires both hunters to work together; catching a hare requires no help at all.

	Stag	Hare
Stag	5, 5	0, 4
Hare	4, 0	4, 4

Hunting stag together is best for both, but it is risky: if your partner defects to hunting hare, you are left with nothing. Hunting hare is safe, since it pays off no matter what your partner does.

Unlike the Prisoners' Dilemma, both “everyone hunts stag” and “everyone hunts hare” will turn out to be self-enforcing once we define equilibrium in Chapter 9.

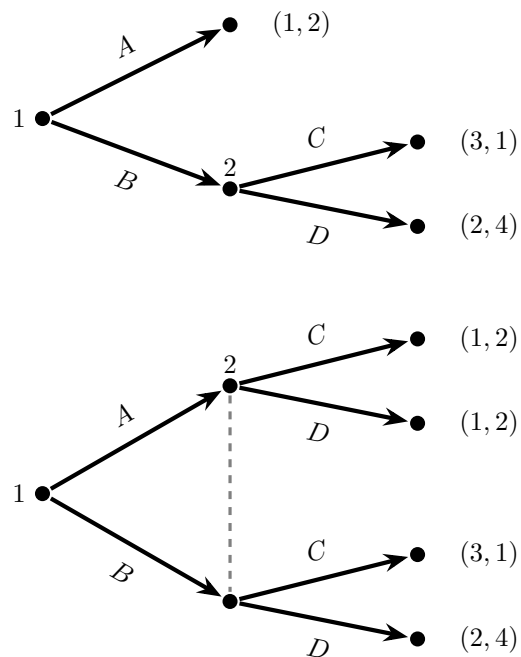
Interpreting the Normal Form

One way to read the normal form is that every player simultaneously and independently commits to a complete contingent plan, then the plans are carried out mechanically.

Converting an extensive-form game into its normal form is always possible, and the result is unique. The reverse is not true: more than one extensive form, even with different information structures, can produce exactly the same normal-form matrix.

When every player moves simultaneously and independently, as in all of this chapter's classic games, there is no timing information to lose. Such games are called **one-shot**, or **static**, games, and the normal form describes them completely.

Example: Same Matrix, Different Trees



	C	D
A	1, 2	1, 2
B	3, 1	2, 4

In the top tree, player 2 only moves after B , so when she moves she knows player 1 did not choose A . In the bottom tree, player 2 always moves, but her single information set spans both arrival nodes, so she never learns whether player 1 chose A or B .

Yet both trees reduce to the same 2×2 matrix.

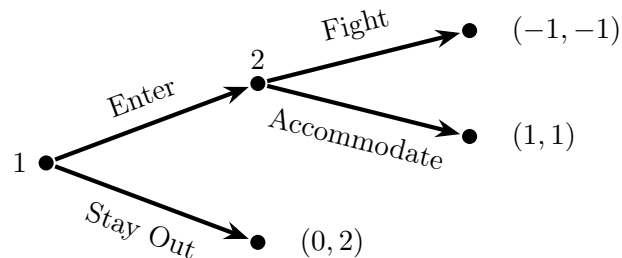
Discussion: A Night Market Entry Game

A new bubble tea stand is deciding whether to Enter a busy night market or Stay Out and set up elsewhere. If it enters, the night market's one established tea stand decides whether to Fight with a price war or Accommodate, keeping prices steady and competing on flavor instead.

- If the newcomer stays out, it earns 0 and the incumbent keeps its usual profit of 2.
- If the newcomer enters and the incumbent fights, a price war leaves both stands with -1 .
- If the newcomer enters and the incumbent accommodates, both stands earn 1.

Draw the extensive form, then write down each stand's strategy space, then construct the normal-form matrix.

Answer: The Night Market Entry Game



The newcomer has one information set with two actions, so

$$S_1 = \{\text{Enter, Stay Out}\}.$$

The incumbent has one information set, reached only if the newcomer enters, so $S_2 = \{\text{Fight, Accommodate}\}$.

The Night Market Matrix

	Fight	Accommodate
Enter	$-1, -1$	$1, 1$
Stay Out	$0, 2$	$0, 2$

Both cells in the “Stay Out” row are identical, for the same reason two rows coincided in the tree-to-matrix example: the incumbent’s plan for a decision it never actually faces cannot change the outcome.

Chapter in a Nutshell

- A strategy is a complete contingent plan: one action for every one of a player's information sets, including those her own plan might never reach.
- A strategy profile lists one strategy per player; the payoff function maps each profile to a payoff for that player.
- The normal form describes a game purely by its strategy spaces and payoff functions, with no reference to timing.
- Converting a tree to its normal form is unique; the reverse is not, since different information structures can still collapse to the same matrix.

Key Concepts

- Strategy, strategy space, and strategy profile describe a player's complete contingent plan and how plans combine across players.
- Payoff function assigns each strategy profile a number for each player, the same numbers that live at the extensive form's terminal nodes.
- Normal form (or strategic form) is the matrix representation of a game, built from strategy spaces and payoff functions alone.
- One-shot, or static, games are games of simultaneous, independent moves, for which the normal form loses no information relative to the extensive form.